Hasan Hüseyin Yurdagül

SENIOR ML ENGINEER · APPLIED ML & LLM SYSTEMS

Adana, Türkiye · Open to EU relocation
Visa sponsorship needed
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PROFILE

Senior Machine Learning Engineer and hands-on Technical Lead with 5+ years building and deploying LLM, NLP, computer vision and recommender systems. Lead a 5-engineer AI team and build retrieval-augmented LLM systems with benchmarked retrieval, fine-tuned open models served on-premises and human review. Architected MdsAI for a municipal support desk; after rollout, median time to first written reply fell from 15.9 h to 2.4 h. Author of Muninn, an open-source Rust agent harness and MCP server.

EXPERIENCE

Universal Software (Universal Yazılım)

Istanbul-based · Remote

Public-sector software vendor; its FlexCity city-management platform runs in 60+ Turkish municipalities.

Technical Lead, AI R&D (hands-on) Jan 2024 – Present

- Lead a 5-engineer AI team, setting the R&D roadmap and mentoring 2 engineers to senior promotion. Wrote the technical work packages for 6 national and Horizon Europe proposals, winning 3 (€1.2M to Universal) and co-leading Teus (river-discharge forecasting).
- Architected MdsAI for the FlexCity support desk: 68% of new tickets get a cited draft from similar solved cases; the rest are re-categorized and routed. Experts approve every reply. After rollout, **median time to first written reply fell from 15.9 h to 2.4 h** (2022–2025 baseline vs. 9,800 tickets in Jan–Jun 2026).
- Built an offline retrieval benchmark: 151 real user questions, each paired with the solved ticket answering it, among 2,000 distractors. With both scoped to the ticket's product, **character n-gram search reached 79.5% Recall@1 vs. 55.6%** for dense MiniLM embeddings, so the simpler method shipped.
- Fine-tuned 7–12B open models (Gemma, Qwen 3) with LoRA and QLoRA on Turkish support tickets. The models now run on company GPUs with vLLM; 61% of generated drafts go out with light or no edits.
- Built a Polars pipeline with embedding clustering (EmbeddingGemma, UMAP, HDBSCAN) so experts could review 79 ticket clusters instead of 29K individual tickets, producing a knowledge base of 15.4K training-grade cases.
- For ticket routing, benchmarked 4 embedding models on 4,500 labeled tickets and chose EmbeddingGemma (0.914 vs. 0.937 for E5-Large, averaged over 3 classifiers, at a quarter of its memory). With support experts, consolidated 194 category labels into 7 queues; after rollout, tickets reassigned after initial routing fell from 24% to 9%.

AI/ML Engineer → Senior AI/ML Engineer (Jan 2023) Apr 2021 – Dec 2023

- Automated field-photo screening for Enerjisa (Türkiye's largest electricity distributor): YOLOv8, small classifiers and OpenCV checks screened 1.2M photos at 1,000+ images/min on one RTX A5000. Flagging 27% as unusable and pre-labeling the rest removed manual triage and **cut annotation work by ~60%**.
- Shipped TF-IDF and embedding-based routing of citizen complaints for client municipalities: 3,000 complaints a month at 85% routing accuracy, with same-day triage instead of 2 days.
- Developed predictive-maintenance classifiers on Enerjisa telemetry that flagged 78% of failures at least 5 days ahead at 2 false alarms per asset per month on a held-out year.
- Built LSTM forecasting pipelines for municipal water demand (İSKİ, Istanbul: 4.1% MAPE at a 7-day horizon vs. 6.3% for the utility's previous method) and industrial gas and electricity consumption.

OxyAI

Remote, Türkiye

AI/ML Engineer (Part-time Contract) May 2021 – Jan 2023

- Designed and shipped recommenders for the online art gallery *Collectors* (text profiles plus CNN visual features; click-through rate rose 14% over the previous popularity list) and the audiobook app *Dinlebi* (120K users; matrix factorization with cold-start onboarding); both stayed in production for 3+ years.

CONSULTING

Freelance ML Consultant (Part-time)

2020 – Present

Alongside employment · Remote, Türkiye

hhyurdagul.com/work

- **Inveon** (e-commerce platform, 2022–2026): built semantic product search (zero-result searches fell 35% across 12 stores after rollout), a production classifier auto-categorizing 20K products a month into 50+ subcategories, and order cancellation and return prediction APIs (IEEE ISAS 2025 paper).
- **Moka** (payment institution, İş Bankası group, since 2023): built and deployed FastAPI services on Azure with Docker (40K calls a month) for merchant-churn scoring, price quoting and revenue forecasting. Churn scores feed monthly retention campaigns; quotes take minutes instead of 2 days by hand.

SELECTED PROJECTS

Muninn – Agent Harness for a Markdown Knowledge Vault

github.com/hhyurdagul/muninn

> Open-source Rust agent that reads, searches and edits a Markdown knowledge vault through a streaming tool-call loop, exposed as a CLI, a TUI and an MCP server. Used daily as shared memory for Claude Code and Codex, delivered in the MCP server instructions with no tool call. Pre-release, in active development.
>
> - The agent loop is tested offline against a scripted mock model server; providers include the native Anthropic API with prompt caching and any OpenAI-compatible endpoint (vLLM, Ollama).
> - Constrained by design: no shell and vault-only file access. Edits to the shared memory and rules files always go through review, stale proposals are rejected, and every change is journaled with conflict-aware undo.

Rust · Tokio · Ratatui · MCP (JSON-RPC) · 7-crate workspace · Nix · Apache-2.0

Fishometry – Single-Image Fish-Length Estimation (MSc thesis)

hhyurdagul.com/projects/fishometry

> Estimates fish length from one photo with the species supplied, without depth sensors or reference objects: YOLO11 landmarks, Depth Anything V2 and SAM segmentation, then DINOv2 features with ridge regression. Reached 5.53 cm MAE vs. 8.76 cm for a species-mean baseline (37% lower) on 348 held-out outdoor photos.

Python · PyTorch · DINOv2 · YOLO11 · SAM · Depth Anything V2 · scikit-learn · Polars

SKILLS

- **LLMs & Agents:** retrieval-augmented generation (RAG), lexical/dense/hybrid retrieval, embeddings, retrieval evaluation, LoRA/QLoRA fine-tuning (Gemma, Qwen), vLLM serving, tool-calling agents, Model Context Protocol (MCP), prompt caching, human-in-the-loop review
- **ML:** PyTorch, scikit-learn, XGBoost/LightGBM, sentence-transformers, UMAP/HDBSCAN; computer vision (YOLO, SAM, DINOv2, OpenCV), recommenders, time-series forecasting (LSTM, seq2seq)
- **Engineering:** Python, Rust, SQL, Polars, FastAPI, Docker, Azure, on-premises GPU serving, MLflow, Weights & Biases, Git, CI, Nix
- **Languages:** Turkish (native), English (C1; IELTS 7.5)

EDUCATION

Çukurova University

Adana, Türkiye

MSc in Computer Engineering — Aug 2023 – Nov 2026 (expected)

BSc in Computer Engineering (English-medium) — Sep 2016 – Aug 2021

PUBLICATIONS

7 peer-reviewed papers (IEEE SIU, ISAS and ISMSIT; IJMCE journal), 3 as first author. Selected:

- *Fault Prediction in Energy Systems: Telemetric Data and Machine Learning Approaches*. IEEE ISAS, 2025. DOI.